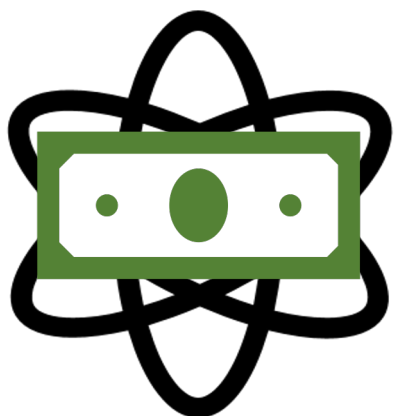
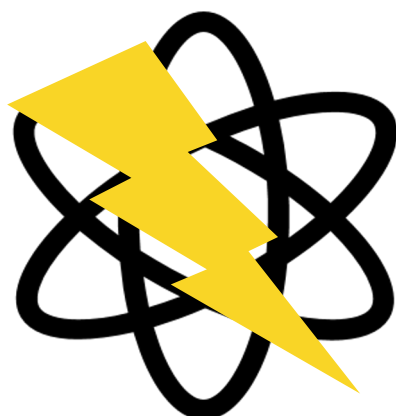


The Status of Quantum Money and Variants

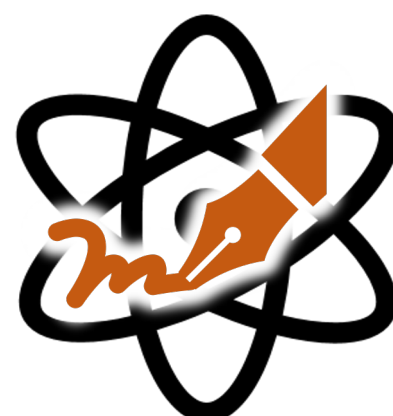
Mark Zhandry
NTT Research



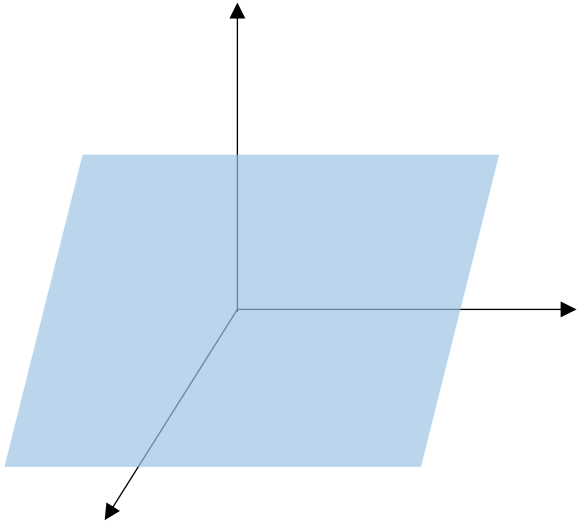
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Hidden Subspaces [Aaronson-Christiano'12]



Provably secure from iO [Z'19]



iO impractical, no other known instantiations



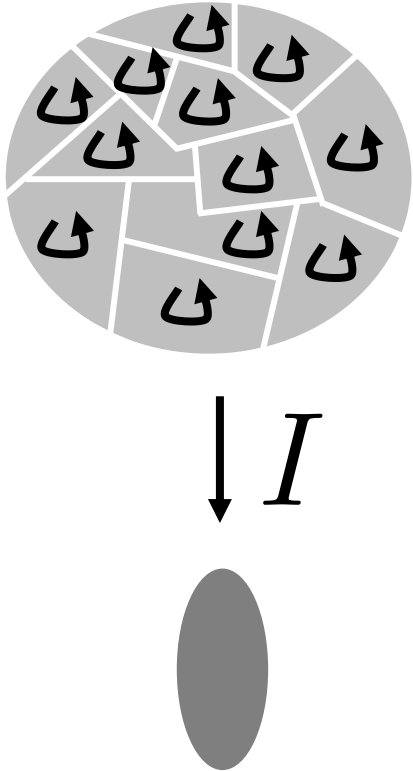
Does not seem amenable to quantum lightning



- Need to know subspace to mint, knowledge of subspace allows for minting many copies

Major open question: (non-evasive) obfuscation of subspaces without iO

Walkable Invariants [Farhi-Gosset-Hassidim-Lutomirski-Shor'10, Liu-Montgomery-Z'23]



Can prove under knowledge assumption



+ statistical assumption [Liu-Montgomery-Z'23]

- Knowledge assumption questionable [Z'24]
- Seems hard to analyze under “nice” assumptions

Practicality unclear



- Knots: what's the security parameter?
- Isogenies: currently incomplete protocol

Readily gives quantum lightning



Does not seem amenable to OSS



- Inherently not collision-resistant

Commuting Unitaries [Kane-Sharif-Silverberg'21]

Only known instantiation (quaternion algebras) needs more study

- Efficiency? Security?

$$U_0 U_1 = U_1 U_0$$

Does not seem amenable to OSS

- Basically no classical structure

?

X

Open question: Find other instantiations

Abelian Group Actions [Z'24]

$$\$ \propto \sum_g e^{i2\pi gh/N} |g * x\rangle$$

Provably secure under “reasonable” assumptions + black box model for isogenies



Should be practical with fault-tolerant QC



Isogenies only known instantiation

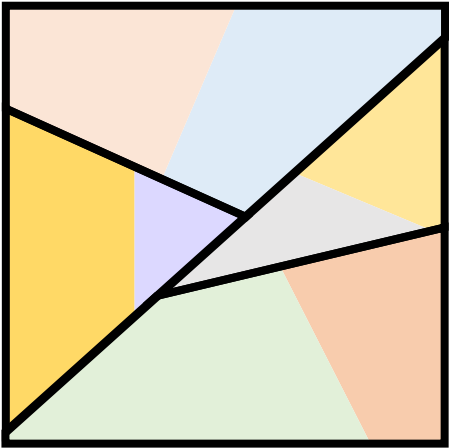


Does not seem amenable to OSS



- Inherently not collision-resistant

“Fractured Subspaces” [Amos-Georgiou-Kiayias-Z’20]



Poor understanding of candidate security



Uses impractical obfuscation



Assuming collision resistance, gives OSS



Open question: Security justification in black-box model

Speculative: Quantum Obfuscation

[Unruh'16] gives quantum oracle relative to which OSS exist

Can we obfuscate this oracle?

Note: existing quantum obfuscation schemes only for classical input/output

[Bartusek-Malavolta'20, Bartusek-Kitagawa-Nishimaki-Yamakawa'23, Bartusek-Brakerski-Vaikuntanathan'14]